

INDU 211 · Comprehensive Topic Guide (Part 5)

Chapter 7: Operations Planning & Control

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Built on the teacher's Lectures 6.0, 7.0 and 8.0 (Chapter 7, parts 1–3), with depth from the course textbook (Hicks, Chapter 7: Operations Planning and Control).

Table of Contents

1. [The Planning and Control Chain](#)
 2. [Operations \(Aggregate\) Planning: Level vs Chase](#)
 3. [Inventory Planning and Control](#)
 4. [The Economic Order Quantity \(EOQ\)](#)
 5. [Material Requirements Planning \(MRP\)](#)
 6. [From MRP to MRP II and ERP](#)
 7. [Scheduling, Dispatching, Lean and JIT](#)
 8. [Demand Forecasting](#)
 9. [Exam Checklist](#)
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1. The Planning and Control Chain

Think of a plant that makes TV sets and radios (Lecture 6.0, slide 3): it buys materials and parts, fabricates components and assembles many configurations, and several products share common components and raw materials. Coordinating all of this is **operations planning and control**.

Operations Planning and Control

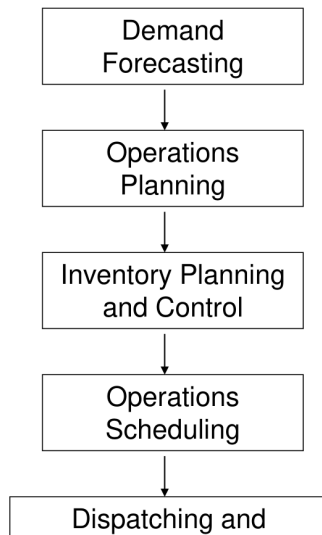


Figure 1: Operations planning and control, Lecture 6.0, slide 4.

Reading the flow, top to bottom: each box feeds the next.

1. **Demand forecasting:** how much will customers want, and when?
2. **Operations planning:** how much capacity (people, overtime, inventory) do we need each period?
3. **Inventory planning and control:** how much of each item to order or make, and when?
4. **Operations scheduling:** which job runs on which machine at what time?
5. **Dispatching and progress control:** release the work to the shop floor, record progress, and correct deviations.

From the textbook (Hicks, §7.2): the chain is hierarchical. Long-range, aggregate decisions constrain the short-range, detailed ones. A good schedule cannot rescue an infeasible operations plan.

2. Operations (Aggregate) Planning: Level vs Chase

Goal (slide 5): have all resources at the right place, at the right time, in the needed quantities, at **minimum cost**, within budget limits and employment practices.

Cumulative Graph

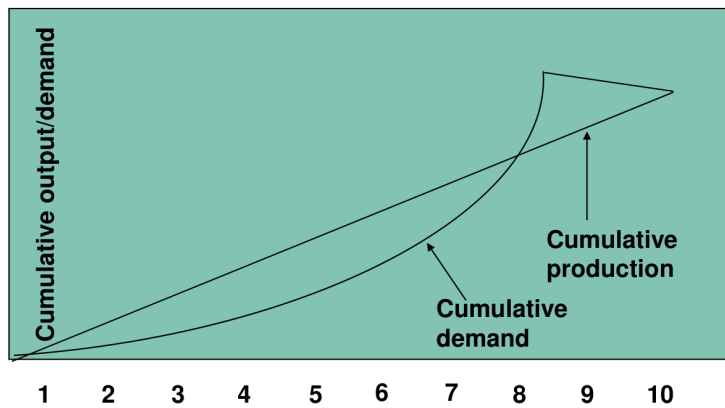


Figure 2: Cumulative graph, Lecture 6.0, slide 6.

Reading the graph: the x-axis is the period (1–10); the y-axis is the running total. The rule is that **cumulative production must never fall below cumulative demand**, otherwise orders are late. The vertical gap between the two curves at any period is the inventory on hand.

Planning Strategies

- Maintain a level workforce
 - Over-time (high demand)
 - Idle-time (low demand)
 - Inventory carrying cost
 - Maintain a steady output rate
- Match demand period by period (chase)
 - Frequent hiring/firing
 - High cost of training
- Use a combination of decision variables
 - The cumulative demand in each period must be satisfied.
 - Plan 1 (level) : high initial inventory, no end inventory
 - Plan 2 (chase): less initial inventory, no end inventory

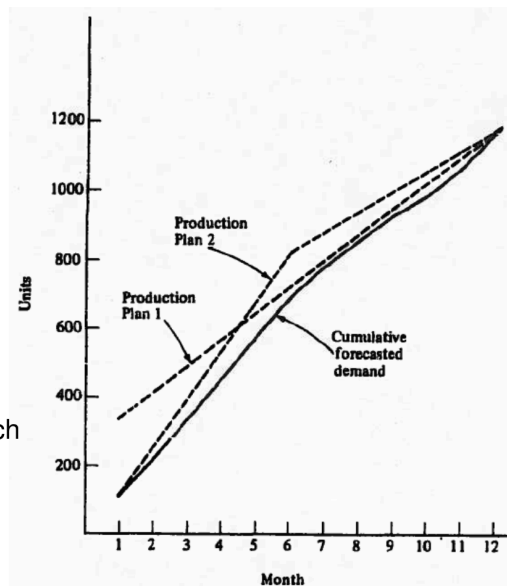


Figure 3: Planning strategies, Lecture 6.0, slide 7.

Strategy	How output is set	Cost it incurs	Inventory
Level (Plan 1)	Steady workforce and output rate	Overtime in peaks, idle time in troughs, inventory carrying cost	High initial inventory, none at the end
Chase (Plan 2)	Output matches demand period by period	Frequent hiring and firing , high training cost	Less inventory
Mixed	Combination of decision variables	Balances both	In between

On the slide's graph the straight line is the level plan (constant slope), while the chase plan hugs the cumulative demand curve.

3. Inventory Planning and Control

Two questions drive every inventory system (slide 8):

- **How much** to order? Too much at once → high carrying cost; too little → too many orders.
- **When** to order? Too often → high ordering cost; too rarely → stock-outs.

Holding inventory...	
Benefits	Better customer service; lower ordering, stock-out, acquisition and quality-related costs; efficient operation
Drawbacks	Higher carrying cost, large-lot quality cost, obsolescence, production-related problems (inventory hides problems)

Cost of inventory (slide 12):

- **Procurement (ordering) cost PC** : the clerical cost of making up and processing an order (fixed per order).
- **Carrying (holding) cost CC** : money tied up, storage space, obsolescence and spoilage, insurance and taxes (per unit per year).

Inventory Systems

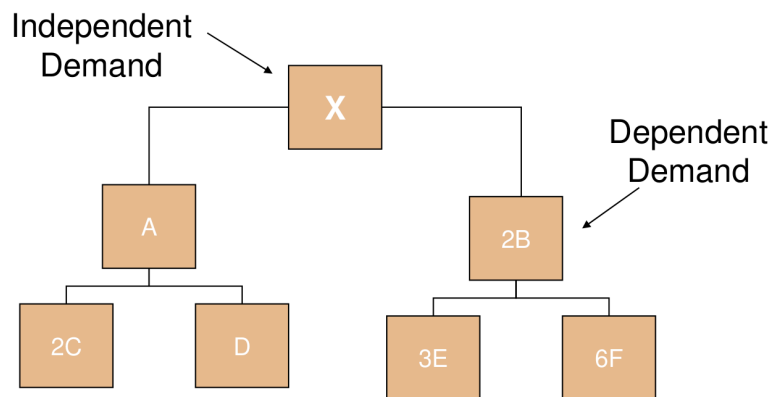


Figure 4: Inventory systems, Lecture 6.0, slide 14.

Reading the tree: X is an **end item**: its demand comes from customers, independent of any other item, so it is forecast and managed with EOQ-type models. A, B, C, D, E, F are components: their demand is **dependent**, calculated from the schedule of X through the tree (2 of C per A, 6 of F per B...). Dependent demand is handled by **MRP** (Section 5).

4. The Economic Order Quantity (EOQ)

Fixed Order Inventory System for Independent Demand

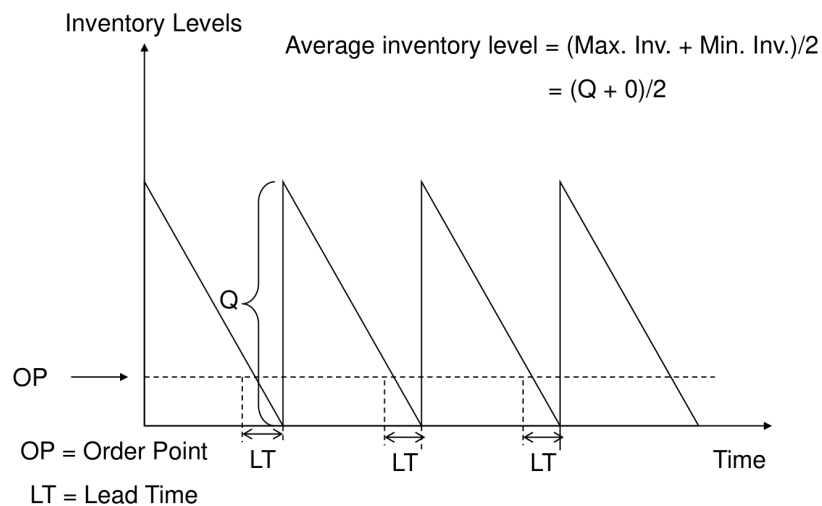


Figure 5: Fixed-order inventory system for independent demand, Lecture 6.0, slide 17.

Reading the saw-tooth: each order of size Q arrives all at once (vertical jump), stock is used at a steady rate (straight descent), and the next order is placed at the **order point (OP)**, one lead time (LT) before stock reaches zero. The average inventory is

$$\text{Average inventory} = \frac{\text{max} + \text{min}}{2} = \frac{Q + 0}{2} = \frac{Q}{2}$$

Assumptions (slides 16 and 18)

Known and constant demand D ; known CC and PC ; orders received all at once; no safety stock; no inventory left when an order arrives; fixed acquisition cost (no quantity discounts).

The cost model

$$TC(Q) = \underbrace{\frac{Q}{2} CC}_{\text{carrying}} + \underbrace{\frac{D}{Q} PC}_{\text{ordering}}$$

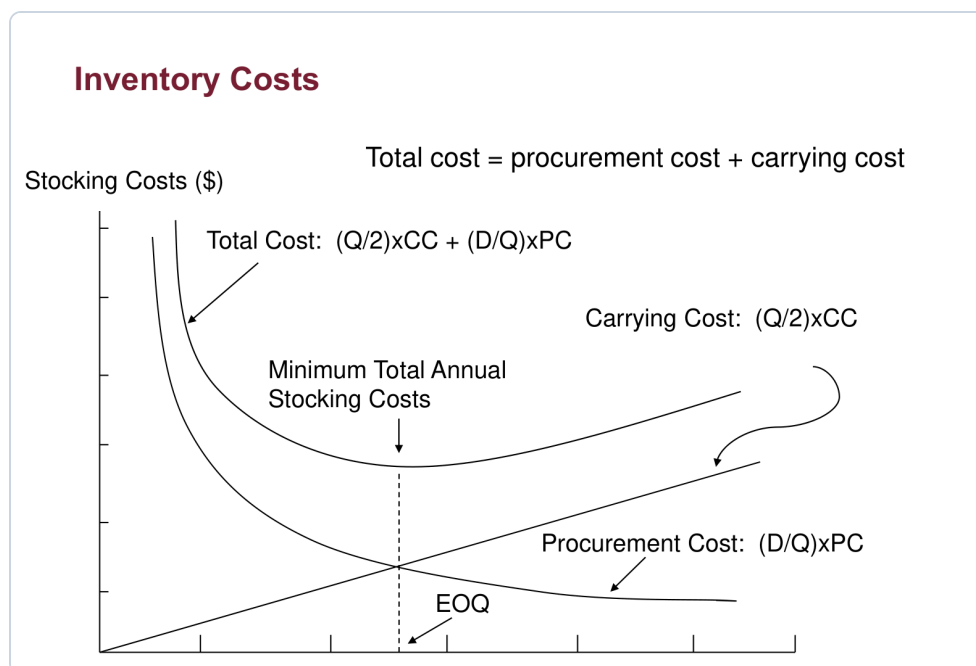


Figure 6: Inventory costs, Lecture 6.0, slide 19.

Reading the curves: the carrying cost is a straight line rising with Q ; the procurement cost falls like $1/Q$ (fewer, larger orders); their sum is U-shaped. The bottom of the U is the EOQ, and it sits exactly where the two cost curves cross.

Deriving the EOQ (slide 20)

$$\frac{dTC}{dQ} = \frac{CC}{2} - \frac{D PC}{Q^2} = 0 \implies Q^2 = \frac{2 D PC}{CC} \implies \boxed{EOQ = \sqrt{\frac{2 D PC}{CC}}}$$

At the EOQ, $\frac{Q}{2}CC = \frac{D}{Q}PC$: annual carrying cost equals annual ordering cost.

Lecture example: Company ABC (slides 21–22)

Example

- Presently, Company ABC uses a rule of thumb to order inventory for widgets. The general manager wonders if the EOQ might save money. The following estimates were calculated from accounting information:

$D = 10,000/\text{year}$

$Q = 400$ per order (present order quantity)

$CC = \$0.40$ per widget per year

$PC = \$5.50$ per order

$$EOQ = \sqrt{\frac{2(5.5)(10000)}{0.4}} = 524.4 \approx 525$$

$$TC = 5.5 \times \frac{10000}{525} + 0.4 \times \frac{525}{2} = 209.76$$

Figure 7: EOQ example, Lecture 6.0, slide 21.

$D = 10,000/\text{year}$, $PC = \$5.50/\text{order}$, $CC = \$0.40/\text{widget}/\text{year}$, current rule $Q = 400$.

$$EOQ = \sqrt{\frac{2(10,000)(5.50)}{0.40}} = \sqrt{275,000} = 524.4 \approx 525$$

Policy	Orders/year	Carrying	Ordering	Total
Current $Q = 400$	25	$\frac{400}{2}(0.40) = \$80$	$25 \times 5.50 = \$137.50$	\$217.50
EOQ $Q = 525$	≈ 19	$\frac{525}{2}(0.40) = \$105$	$\frac{10,000}{525}(5.50) = \104.76	\$209.76

Switching saves about \$7.74 per year here. Small, but the same logic applied to thousands of items is significant.

EOQ model variations (slide 23): stock-outs and responsiveness, quantity discounts, demand variation, gradual inventory build-up. Each needs a revised model.

5. Material Requirements Planning (MRP)

MRP (slide 3) is a computer-based technique that determines the components needed to meet a known **master schedule** for end items, minimising inventory while keeping enough material for production.

Materials Requirement Planning - MRP

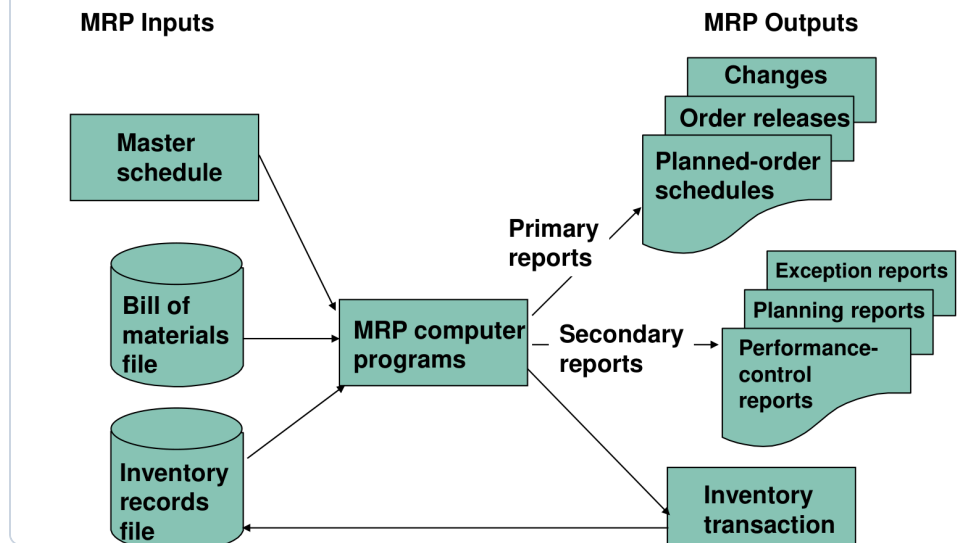


Figure 8: MRP inputs and outputs, Lecture 7.0, slide 4.

Reading the diagram: three input files, the **master schedule** (what end items, when), the **bill of materials** (what each is made of) and the **inventory records** (what is on hand and on order), feed the MRP programs. They produce **primary reports** (planned-order schedules, order releases, changes) and **secondary reports** (exception, planning and performance reports).

Product Structure Tree

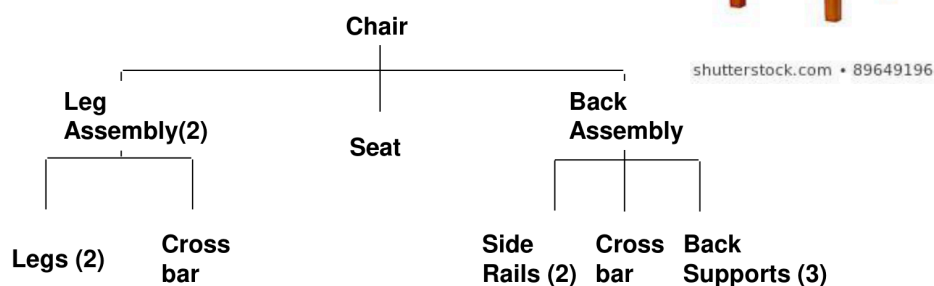


Figure 9: Product structure tree, Lecture 7.0, slide 5.

Reading the tree: one chair needs **2 leg assemblies**, 1 seat and 1 back assembly. Each leg assembly needs 2 legs and 1 cross bar; the back assembly needs 2 side rails, 1 cross bar and 3 back supports. So 100 chairs need $100 \times 2 \times 2 = 400$ legs and $100 \times (2 \times 1 + 1) = 300$ cross bars. MRP "explodes" the end-item quantity down the tree in exactly this way.

The MRP record (one per item)

Row	Meaning
Gross requirements	Total needed each week (from the parent's planned-order releases × quantity per parent)
Scheduled receipts	Orders already placed, arriving that week
Projected on hand	Stock carried forward
Net requirements	Gross – (on hand + scheduled receipts), if positive
Planned-order receipts	Net requirement rounded up to the lot size
Planned-order releases	Planned receipts shifted earlier by the lead time

Lecture Example 1: product P1 (slides 7–8)

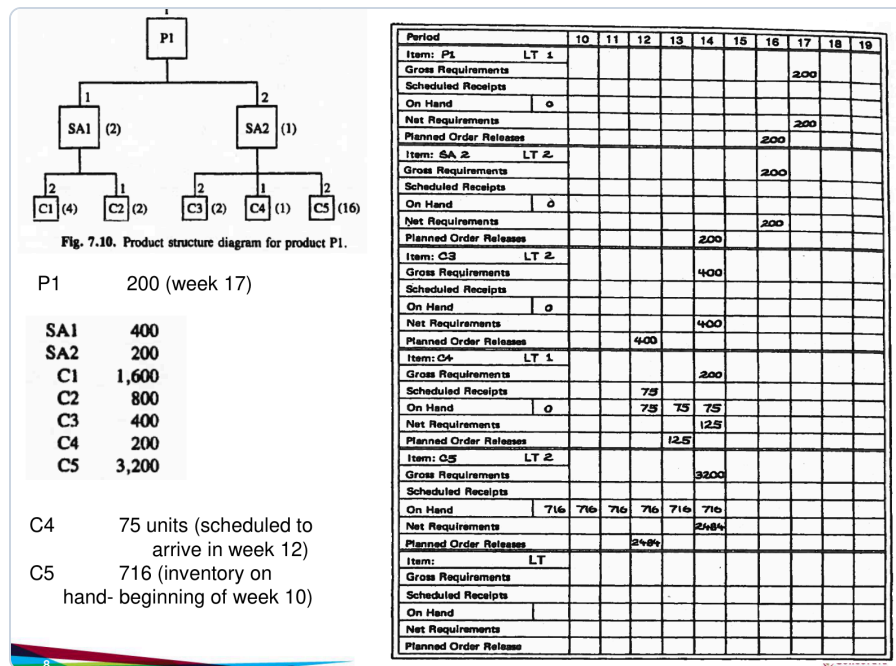


Figure 10: MRP Example 1 records, Lecture 7.0, slide 8.

Reading the records for C5 (16 per SA2, lead time 2, 716 on hand):

Gross (week 14) = $200 \times 1 \times 16 = 3,200$, Net = $3,200 - 716 = 2,484$, release in week 14 –

And **C4** (1 per SA2, LT 1, 75 arriving in week 12): gross 200 in week 14, net $200 - 75 = 125$, released in week 13.

Lecture Example 2: shutters with lot sizes (slides 9–10)

Master Schedule		Week Number	1	2	3	4	5	6	7	8
for Shutters:		Quantity	100							150
Shutters:		Gross requirements	0	0	0	100	0	0	0	150
		Scheduled receipts								
Quantity =	1	Projected on hand	0	0	0	0	0	0	0	0
LT =	1	Net requirements	0	0	0	100	0	0	0	150
Lot Size =	1	Planned-order receipts	0	0	0	100	0	0	0	150
		Planned-order releases	0	0	100	0	0	0	150	0
Frames:		Gross requirements	0	0	200	0	0	0	300	0
		Scheduled receipts								
Quantity =	2	Projected on hand	0	0	0	120	120	120	120	140
LT =	2	Net requirements	0	0	200	0	0	0	180	0
Lot Size =	320	Planned-order receipts	0	0	320	0	0	0	320	0
		Planned-order releases	320	0	0	0	320	0	0	0
Wood sections:		Gross requirements	0	0	400	0	0	0	600	0
		Scheduled receipts	70							
Quantity =	4	Projected on hand	70	70	70	20	20	20	20	50
LT =	1	Net requirements	0	0	330	0	0	0	580	0
Lot Size =	70	Planned-order receipts	0	0	350	0	0	0	630	0
		Planned-order releases	0	350	0	0	0	630	0	0

Note: Use Lot Size = 1 for lot-for-lot ordering.

Figure 11: MRP Example 2 (shutters, frames, wood sections), Lecture 7.0, slide 10.

Reading the frames record (2 per shutter, LT 2, lots of 320): shutter releases of 100 (week 3) and 150 (week 7) create frame gross requirements of 200 and 300. Week 3: net 200 → receive one lot of 320, 120 left. Week 7: net 300 – 120 = 180 → another lot of 320, 140 left. Releases: **320 in week 1 and 320 in week 5**. The wood sections (4 per shutter, lots of 70, 70 already scheduled) follow the same logic: releases of 350 in week 2 and 630 in week 6.

6. From MRP to MRP II and ERP

Stage	What it adds
MRP	Component requirements from the master schedule
Closed-loop MRP	Feedback: completed production and stock on hand flow back so plans can be checked against capacity and adjusted
MRP II (manufacturing resource planning)	Integration with marketing, finance and purchasing, plus a simulation component for what-if questions
ERP (enterprise resource planning)	An MRP II system that ties customer orders to enterprise-wide resources and suppliers in one database and interface

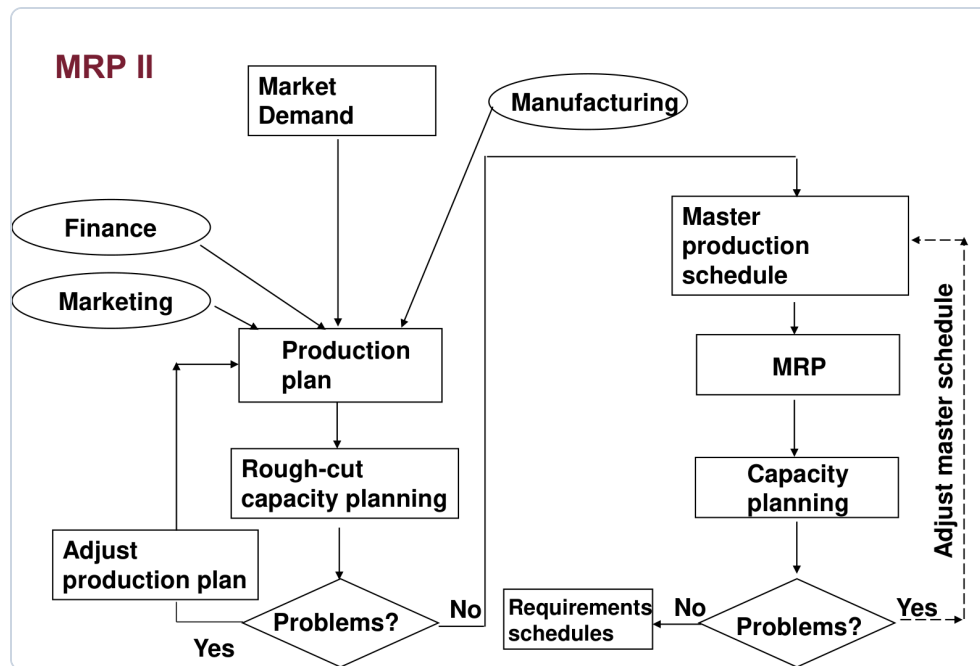


Figure 12: MRP II, Lecture 7.0, slide 14.

Reading the flow: market demand, finance, marketing and manufacturing together set the **production plan**; **rough-cut capacity planning** checks it. If there are problems, the plan is adjusted (left loop). Then the **master production schedule** drives **MRP**, and detailed **capacity planning** checks it again; problems adjust the master schedule (right loop). The two feedback loops are what "closed loop" means.

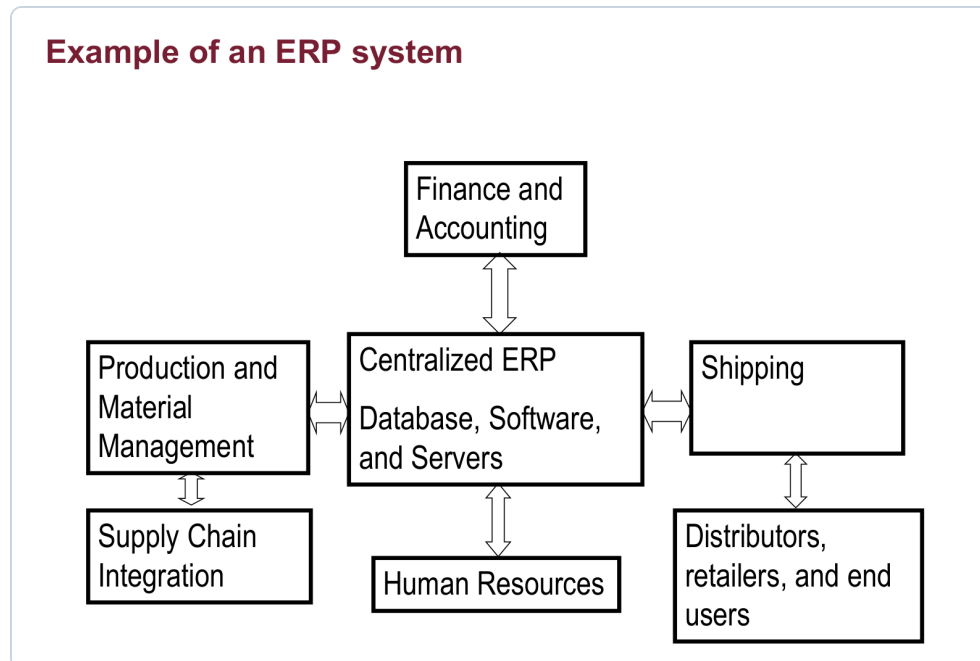


Figure 13: Example of an ERP system, Lecture 7.0, slide 16.

ERP benefits: integrated systems across locations, less inventory, faster response and delivery. **Drawbacks:** complex, expensive (average total cost of ownership \$15 million on slide 18), long implementations (1–3 years), technology-dependent.

7. Scheduling, Dispatching, Lean and JIT

Operations scheduling

- Assign specific operations to specific operating facilities with specific start and end times
- Minimized cost
 - WIP inventory
 - Idle operators
 - Idle equipment
 - Over-time
 - Lateness
- Scarce resources (machines)
- External random perturbations
 - Canceled/emergency order
 - Late delivery of material
 - Machine/tool breakdown

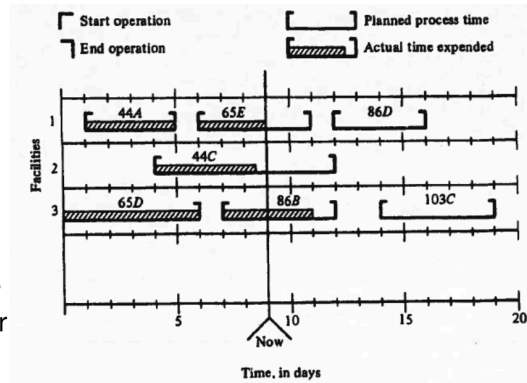


Fig. 7.9. Gantt scheduling chart.

Figure 14: Operations scheduling, Lecture 7.0, slide 19.

Operations scheduling assigns operations to facilities with start and end times, minimising WIP, idle operators and equipment, overtime and lateness, despite machine breakdowns, emergency orders and late material. The Gantt chart shows each machine as a row and each job as a bar, with planned vs actual progress.

Dispatching and progress control turns plans into action: relay the schedule to supervisors, record progress, analyse deviations and take corrective action.

Lean manufacturing and the seven wastes

Waste (*muda*) is anything beyond the minimum equipment, materials, parts, space and worker time that add value (Shoichiro Toyoda). The seven wastes: **overproduction, waiting, transportation, inefficient processing, inventory, unnecessary motion, product defects.**

Just-in-Time (JIT)

- A philosophy of continuous, forced problem solving: material is **pulled** through the system to arrive exactly when needed.
- **Push** (MRP-style): material is released downstream regardless of whether the next station is ready. **Pull** (JIT): a station works only when the downstream station signals it needs material.
- JIT attacks waste, **exposes problems** caused by variability, and streamlines production by reducing inventory.

Just-in-Time Success Factors

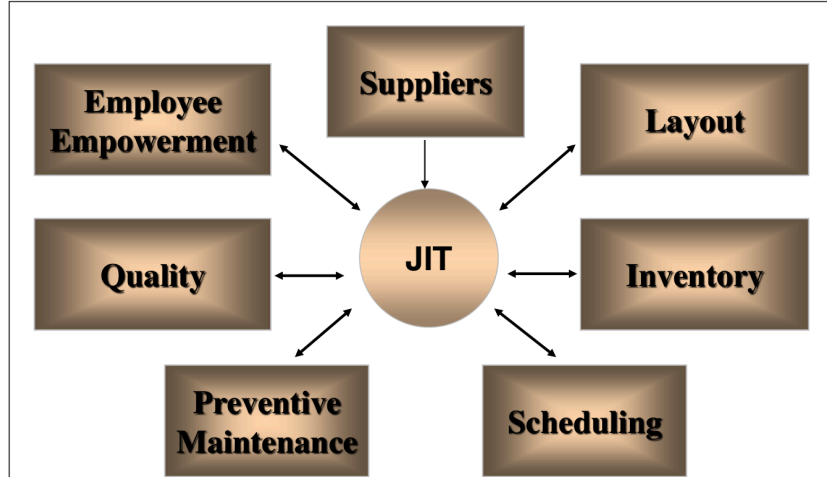


Figure 15: Just-in-time success factors, Lecture 7.0, slide 27.

Kanban system

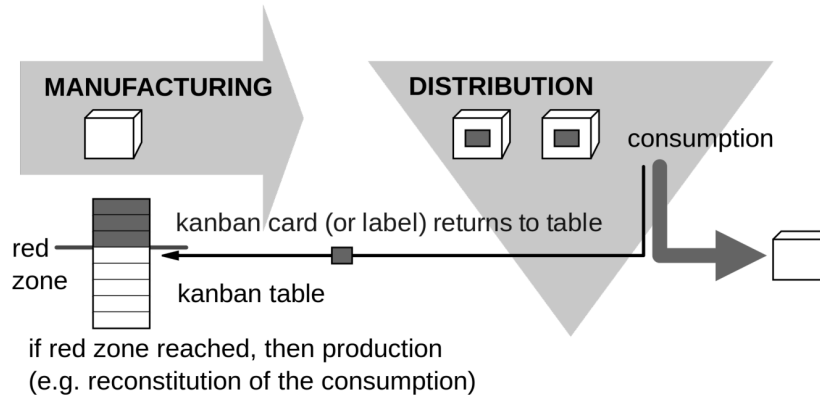


Figure 16: Kanban system, Lecture 7.0, slide 29.

Reading the Kanban figure: when distribution consumes a container, its **kanban card** returns to the **kanban table** at manufacturing. Cards accumulating in the **red zone** trigger production. So the **rate of consumption controls the rate of production**, and adding or removing containers (cards) changes the production rate.

MRP	Kanban
Computed plan, "optimal" if followed	Not aiming at an optimum
Top-down (push)	Bottom-up (pull)

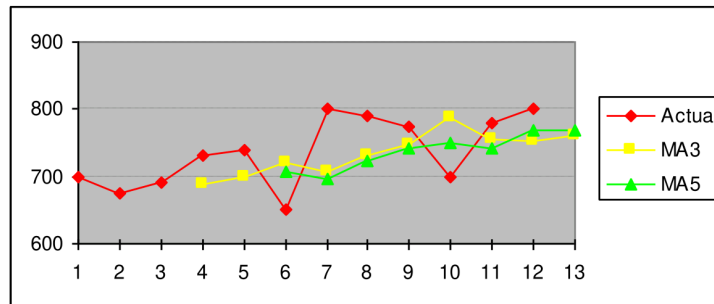
8. Demand Forecasting

Principles (Lecture 8.0, slide 3): forecasting assumes the future is related to the past; forecasts contain error; group forecasts are more accurate than individual ones; accuracy decreases as the horizon lengthens. Forecasts are either **judgmental** (executive opinion, surveys) or **time series** (trend, seasonality, irregular and random variation).

Simple moving average

$$\hat{x}_t = \frac{1}{n} \sum_{i=1}^n x_{t-i}$$

Simple Moving Average



The next period's demand: the average of the actual demand values for the last n time periods

$$\hat{x}_t = \frac{1}{n} \sum_{i=1}^n x_{t-i}$$

Figure 17: Simple moving average, Lecture 8.0, slide 8.

Reading the chart: MA3 (3 periods) follows the actual demand more closely but is noisier; MA5 is smoother but lags turning points. **Larger n = more smoothing, slower response.**

Lecture example (slide 9): demands 42, 40, 43, 40, 41 give $\hat{x}_6 = \frac{43+40+41}{3} = 41.3$.

Exponentially weighted moving average (EWMA)

$$\hat{x}_t = \hat{x}_{t-1} + \alpha (x_{t-1} - \hat{x}_{t-1}) = \alpha x_{t-1} + (1 - \alpha) \hat{x}_{t-1}$$

Example: Prepare a forecast using a smoothing constant of 0.40

Period	Number of complaints	Forecast	Calculations
1	60		
2	65	60	Use previous value of series
3	55	62	$60 + 0.4(65-60)=62$
4	58	59.2	$62 + 0.4(55-62)=59.2$
5	64	58.72	$59.2 + 0.4(58-59.2)=58.72$
		60.83	$58.72+0.4(64-58.72)=60.83$

Figure 18: EWMA example with $\alpha = 0.40$, Lecture 8.0, slide 11.

Period	Actual	Forecast	Calculation
1	60	—	
2	65	60	first forecast = previous actual
3	55	62	$60 + 0.4(65 - 60)$
4	58	59.2	$62 + 0.4(55 - 62)$
5	64	58.72	$59.2 + 0.4(58 - 59.2)$
6		60.83	$58.72 + 0.4(64 - 58.72)$

Impact of the smoothing factor on the forecast

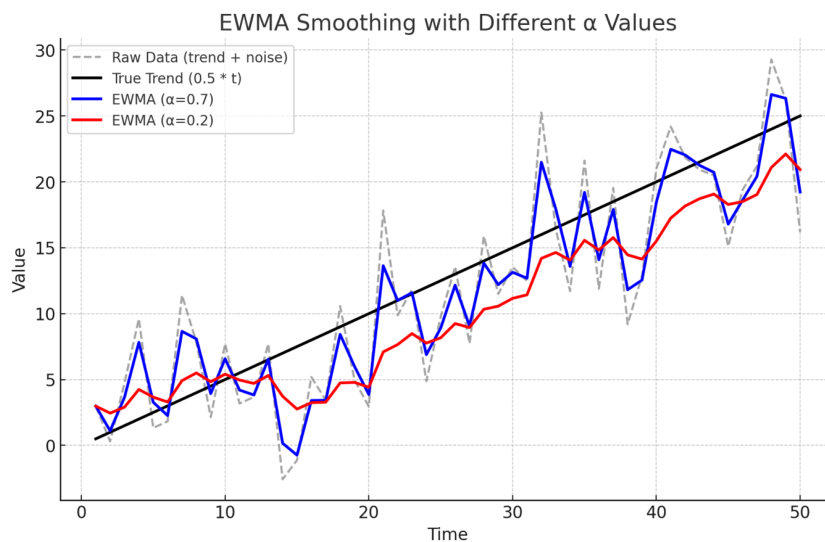


Figure 19: Impact of the smoothing factor, Lecture 8.0, slide 12.

Reading the chart: a large α tracks the data closely (responsive but noisy); a small α gives a smooth curve that lags. With $\alpha = 0.2$, the latest observation gets 20% of the weight and the remaining 80% decays exponentially across all older data.

Linear regression (trend)

$$\hat{x}_t = a + bt, \quad b = \frac{n \sum tx - \sum t \sum x}{n \sum t^2 - (\sum t)^2}, \quad a = \frac{\sum x - b \sum t}{n}$$

Linear regression example

t Week	t ²	x Sales	tx
1	1	150	150
2	4	157	314
3	9	162	486
4	16	166	664
5	25	177	885
$\sum t = 15$ $(\sum t)^2 = 225$	$\sum t^2 = 55$	$\sum x = 812$	$\sum tx = 2499$

Figure 20: Linear regression example, Lecture 8.0, slide 16.

$$b = \frac{5(2499) - 15(812)}{5(55) - 225} = \frac{315}{50} = 6.3, \quad a = \frac{812 - 6.3(15)}{5} = 143.5, \quad \hat{x} = 143.5 + 6.3t$$

Forecast for week 6: $143.5 + 6.3(6) = 181.3$. **Watch the denominator:** $\sum t^2 = 55$ is not $(\sum t)^2 = 225$.

9. Exam Checklist

- [] Order of the chain: forecast → operations plan → inventory → schedule → dispatch.
- [] Level vs chase: which cost each strategy incurs.
- [] Independent (end items, EOQ) vs dependent (components, MRP) demand.
- [] EOQ formula, derivation, $Q/2$ average inventory, carrying = ordering at the optimum.
- [] MRP record: gross → net → lot size → **offset by lead time**.
- [] MRP → closed loop → MRP II → ERP: what each adds.
- [] Seven wastes; push vs pull; how Kanban controls production.

- [] Moving average (most recent n), EWMA (α multiplies the error), regression slope and intercept.